

Advanced Natural Language Processing

The Problem with Natural Language

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Maastricht University

Learning Objectives

- Identify the main sources of ambiguity in natural language (lexical, syntactic, semantic, pragmatic)
- Explain why rule-based NLP systems struggle with real-world text
- Explain Zipf's Law and why it makes the “long tail” of language unsolvable by fixed rules
- Describe how the classical NLP pipeline (morphology → syntax → semantics → pragmatics) re-emerges inside modern Transformer models

Ambiguity in Natural Language Processing

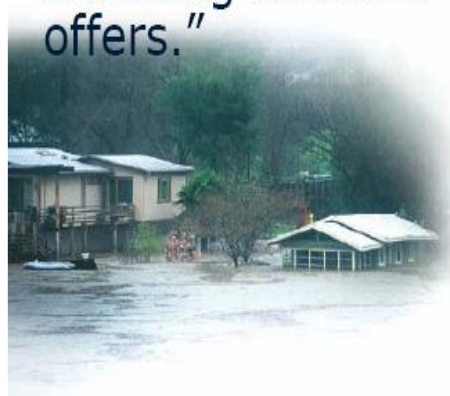
Linguistic Ambiguity:

natural language can have more than one meaning.

Disambiguating such linguistic structures can be complex and is one of the main challenges of the field of natural language processing.

Understanding Language is Essential for Understanding Context

"**Bank of California floods** mailboxes with checking account offers."

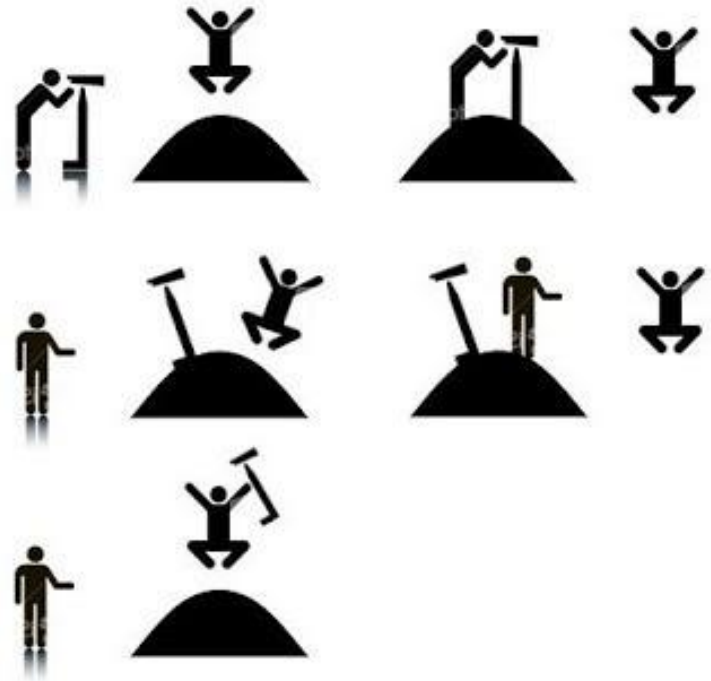


"**Banks of California** river breached in **flood**."

Teaching note: Ambiguity has many flavours (lexical, syntactic, semantic) — defined a few slides on. → ambiguity requires context beyond the sentence

Ambiguities - Examples

- “I saw the man on the hill with a telescope.”
- I saw the man. The man was on the hill. I was using a telescope.
- I saw the man. I was on the hill. I was using a telescope.
- I saw the man. The man was on the hill. The hill had a telescope.
- I saw the man. I was on the hill. The hill had a telescope.
- I saw the man. The man was on the hill. I saw him using a telescope.



Ambiguities - Jokes

- One morning I shot an elephant in my pajamas. How he got into my pajamas, I'll never know.
- Policeman to little boy: "We are looking for a thief with a bicycle."
Little boy: "Wouldn't you be better using your eyes."
- Why is the teacher wearing sunglasses. Because the class is so bright.



Teaching note: All 3 jokes exploit PP-attachment or lexical ambiguity — spot which is which.

Other problems we need to deal with

Contextual words and phrases and homonyms:

- *I **ran** to the store because we **ran** out of milk.*
- *Can I **run** something past you really quick?*
- *The house is looking really **run** down.*
- *The hospital took some blood **cells** from the inmate in his **cell**. The results were transmitted by using a **cell** phone.*

synonyms:

- small, little, tiny, minute

Complex Semantic Relations — Examples

- **Synonyms:** happy / joyful
- **Antonyms:** hot $\leftrightarrow$ cold
- **Hyponymy:** “rose” is a hyponym of “flower”
- **Homonyms:** “bank” (river) / “bank” (money)
- **Meronymy:** “wheel” is part of “car”
- **Metonymy:** “The White House announced...” (place stands for institution)
- **Retronymy:** “acoustic guitar” (coined after “electric guitar” existed)
- **Polysemy:** “paper” (material / academic article / newspaper)
- **Converseness:** buy $\leftrightarrow$ sell, parent $\leftrightarrow$ child

Takeaway: these relations are exactly what WordNet encodes by hand — and what embeddings later learn automatically from data.



Irony and Sarcasm



Mailman Rob @kingrob619 · Nov 12

...

Replying to @JakeMillard4 @Walmart and @PlayStation

Its 🏀. @Sony and @PlayStation said this would be the most accessible console of them all. Yeah right.

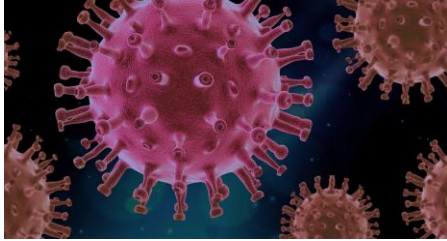


Ambiguity

Teaching note: Compare to the telescope & elephant examples earlier — same 3 categories. Irony/sarcasm is pragmatic: literal and intended meaning diverge — harder than lexical/syntactic/semantic ambiguity.

- Lexical ambiguity:** a word that could be used as a verb, noun, or adjective.
- Semantic ambiguity:** the interpretation of a sentence in context. For example: *I saw the boy on the beach with my binoculars.* This could mean that I saw a boy through my binoculars, or the boy had my binoculars with him
- Syntactic ambiguity:** In the sentence above, this is what creates the confusion of meaning. The phrase *with my binoculars* could modify the verb, “saw,” or the noun, “boy.”

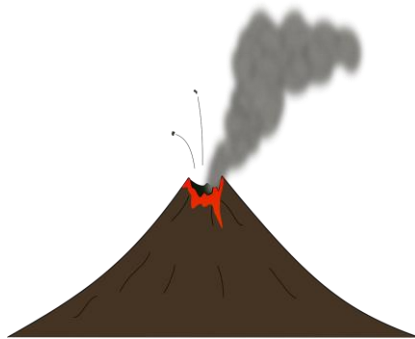
Living language



Covid



Brexit

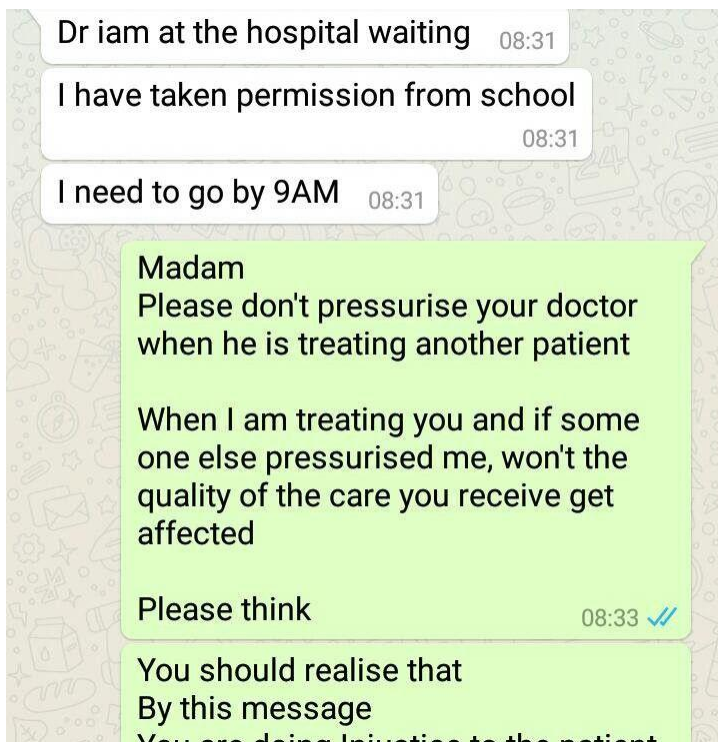


Eyjafjallajökull



face mask
Mondkapje

Errors, Incomplete Sentences and Slang



Twitter is my alter eagle

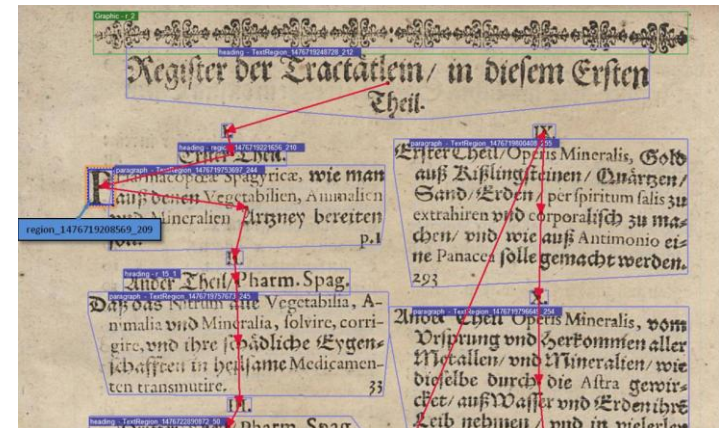
◀ Reply ↻ Retweet ★ Favorite



Teaching note: Typos, phonetic spelling ("alter eagle"), garbled names — real text is noisy by default.

Scanning (OCR) Errors

At the opening of this Court the Gouvernor¹ declard it, yt on the certeine or generall rumors in M^r Jenner, lately arrived, yt O^r charter was condemned, & judgment entred vp, &c, they lookt at it as an incumbent duty to acquaint the Court wth it, & leaue the consideration of what was or might be necessary to them, &c.²



Teaching note: OCR confuses similar glyphs (long-s→f, rn→m). Noise can enter before the NLP system ever sees “clean” text.

PASTORALISTS AND THE SHEARERS' UNIONS.	THE PASTORALISTS AND THE SHEARERS' UNIONS.	THE PASTORALISTS AND SHEARERS' UNION
is (says the <i>Age</i> of Tuesday last) being instituted amongst members of the Union in consequence of a number of communications having been received by station-owners from shearers in whom expressing the reluctance they had struck work, and in which they had at the outset been forced union against their will, and in ulterior consequences threatened of their refusal. Furthermore, it the knowledge of the Pastoralists' Union that many of these shearers of the Shearers' Union use they laboured under the impression if they refused to do so they would be liable to have fines and penalties imposed on them by an action at law. So	An inquiry is (says the <i>Age</i> of Tuesday last) being instituted amongst members of the Pastoralists' Union in consequence of a number of communications having been received by individual station owners from shearers in their employment expressing the reluctance with which they had struck work, and in which they had at the outset been forced to join the union against their will, and in dread of the ulterior consequences threatened in the event of their refusal. Furthermore, it has come to the knowledge of the Pastoralists' Union that many of these shearers objected to the Shearers' Union to stoke because they laboured under the impression that if they refused to do so they would be liable to have fines and penalties imposed on them by an action at law.	An inquiry is (says the <i>Age</i> of Tuesday last) being instituted amongst members of the Pastoralists' Union in consequence of a number of communications having been received by individual station owners from shearers in their employment expressing the reluctance with which they had struck work, and in which they had at the outset been forced to join the union against their will, and in ulterior consequences threatened of their refusal. Furthermore, it has come to the knowledge of the Pastoralists' Union that many of these shearers objected to the Shearers' Union to stoke because they laboured under the impression that if they refused to do so they would be liable to have fines and penalties imposed on them by an action at law.

OCR = Optical Character Recognition. We see these errors a lot in historical archives that are put on-line and in text from bitmap (scanned) PDF's.

Different Languages and domain dependent language

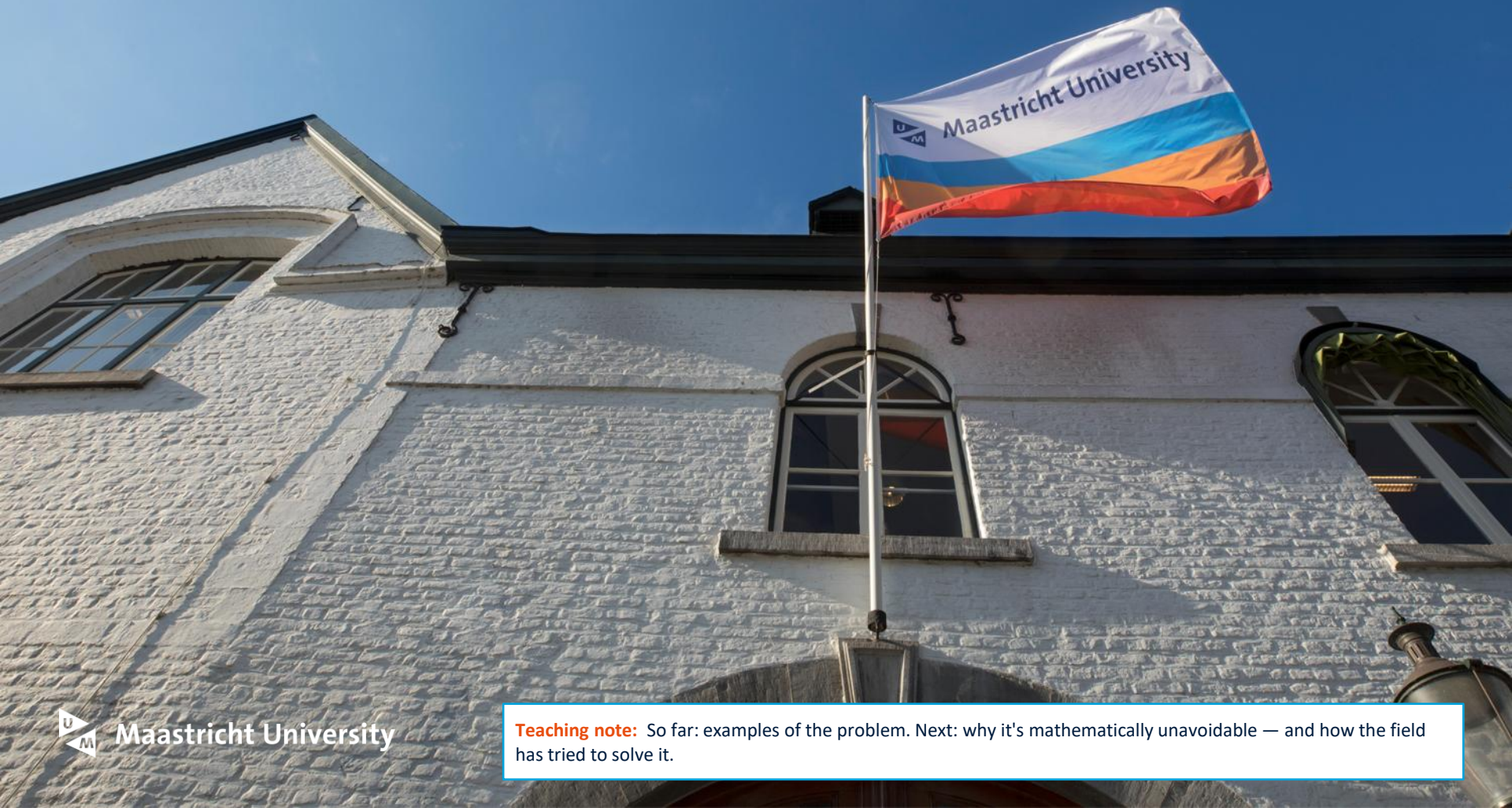


Teaching note: Two issues bundled here: multilinguality (scripts, word order) and domain register (signage vs. advertising vs. legal/medical).
A model trained on one language/domain doesn't automatically transfer to another.

And there is much more to deal with ...

- Discourse: the red line of a narrative
- Intent, actions, goals, ...
- Relation between a textual reference and an object in the real world
- Hidden meaning or bias in relation to language.
- Jokes, speeches, ...
- Morality, culture, boundaries, politics, background, education, ...

Let's take a closer look at “natural language”



Language is the Perfect Example of a “long Tail”



Zipf's law was originally formulated in terms of quantitative linguistics, stating that given some corpus of natural language utterances, the frequency of any word is inversely proportional to its rank in the frequency table. Thus the most frequent word will occur approximately twice as often as the second most frequent word, three times as often as the third most frequent word, etc. -

https://en.wikipedia.org/wiki/Zipf%27s_law

Zipf's law is even more dominant for idiom, proverbs and domain dependent meanings.

In statistics, a long tail of some distributions of numbers is the portion of the distribution having many occurrences far from the "head" or central part of the distribution. **Long tails are impossible to model with rules-based models as one will suffer significantly from cases that are not covered by the model.**

Co-references & Pronouns



Dr. Macklin often brings his dog Champion to visit with the patients. **He** just loves to give big, wet, sloppy kisses!

„If the baby does not like the milk, boil it“

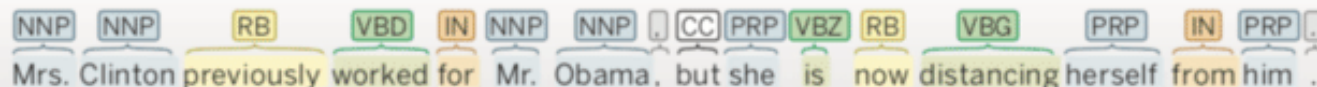


impossible
impatient
unsafe
unknown
unemployed

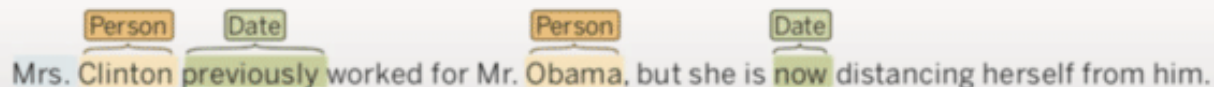
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(Long and short) Distance Complex Relations

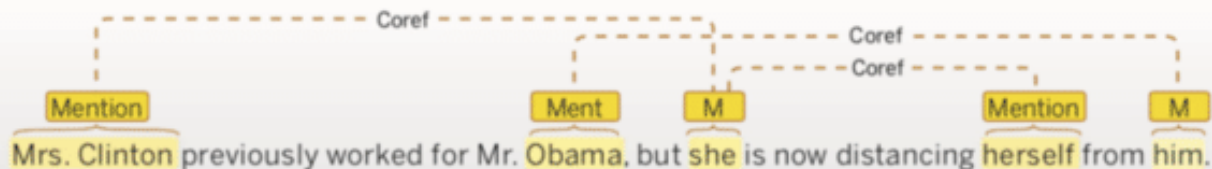
Part of speech:



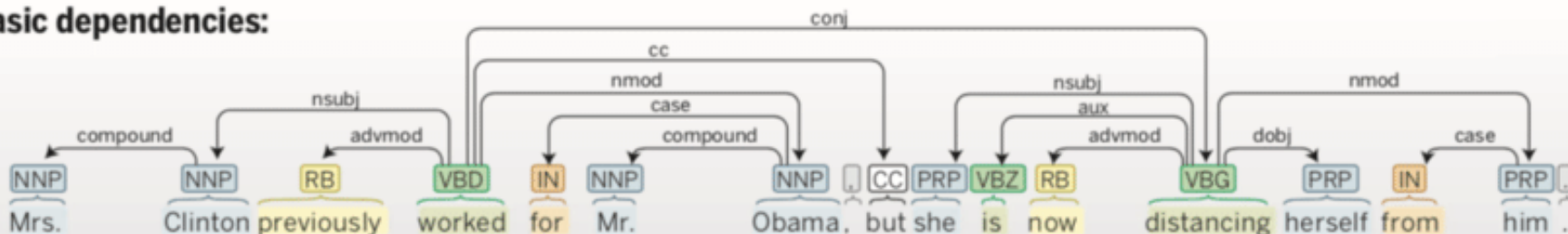
Named entity recognition:



Co-reference:



Basic dependencies:



In this course we will discuss various methods to deal with these linguistic phenomena

Key NLP Task Definitions

- **POS tagging:** label each word's grammatical category
- **NER:** identify names of people, places, dates, organizations
- **Coreference:** link “she” / “her” / “him” back to the entity they refer to
- **Dependency parsing:** identify grammatical relations between words (subject, object, modifier)

Takeaway: these four tasks are the building blocks of the classical NLP pipeline — and reappear later as BERT's internal layers.



Different Approaches to address these issue in NLP we will discuss in this course:

Grammar (rules) and logic-based Models

— *hand-written rules; precise but brittle*

Probabilistic Models

— *adds likelihoods to handle ambiguity*

Corpus-based Models

— *learns patterns directly from annotated data*

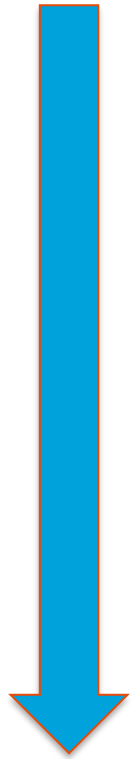
Early ANN and Deep Learning (RNN and LSTM)

— *learns sequential dependencies automatically*

Transformers

— *self-attention scales to long-distance dependencies*

Approaches to NLP



- Generative Grammars (rules & logic)
- Probabilistic models
- Corpus Linguistics, statistics & discriminative models
- Early Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) models, and Gated Recurrent Units (GRU).
- Transformers

And all possible combinations

This arrow spans ~50 years of NLP research, each generation attacking ambiguity + the long tail.

Natural Language ...

- **Natural Language Processing (NLP)** is a field of computer science and linguistics concerned with the interactions between computers and human (natural) languages.
- **Natural Language Generation (NLG aka syntheses)** systems convert information from computer databases into readable human language.
- **Natural Language Understanding (NLU)** systems convert samples of human language into more formal representations such as parse trees or first-order logic structures that are easier for computer programs to manipulate.
- **Natural Language Inference (NLI)** is the task of determining whether the given "hypothesis" logically follows from the "premise". In layman's terms, you need to understand whether the hypothesis is true, while the premise is your only knowledge about the subject.

Many problems within NLP apply to both generation and understanding; for example, a computer must be able to model **morphology** (the structure of words) in order to understand an English sentence or paragraph.

Let's try to identify these individual words first.... this is what we call "tokenization".

Teaching note: E.g., NLG = weather report from data; NLU = parsing "book me a flight" into intent+slots; NLI = does "penguins can fly" follow from "all birds can fly"?

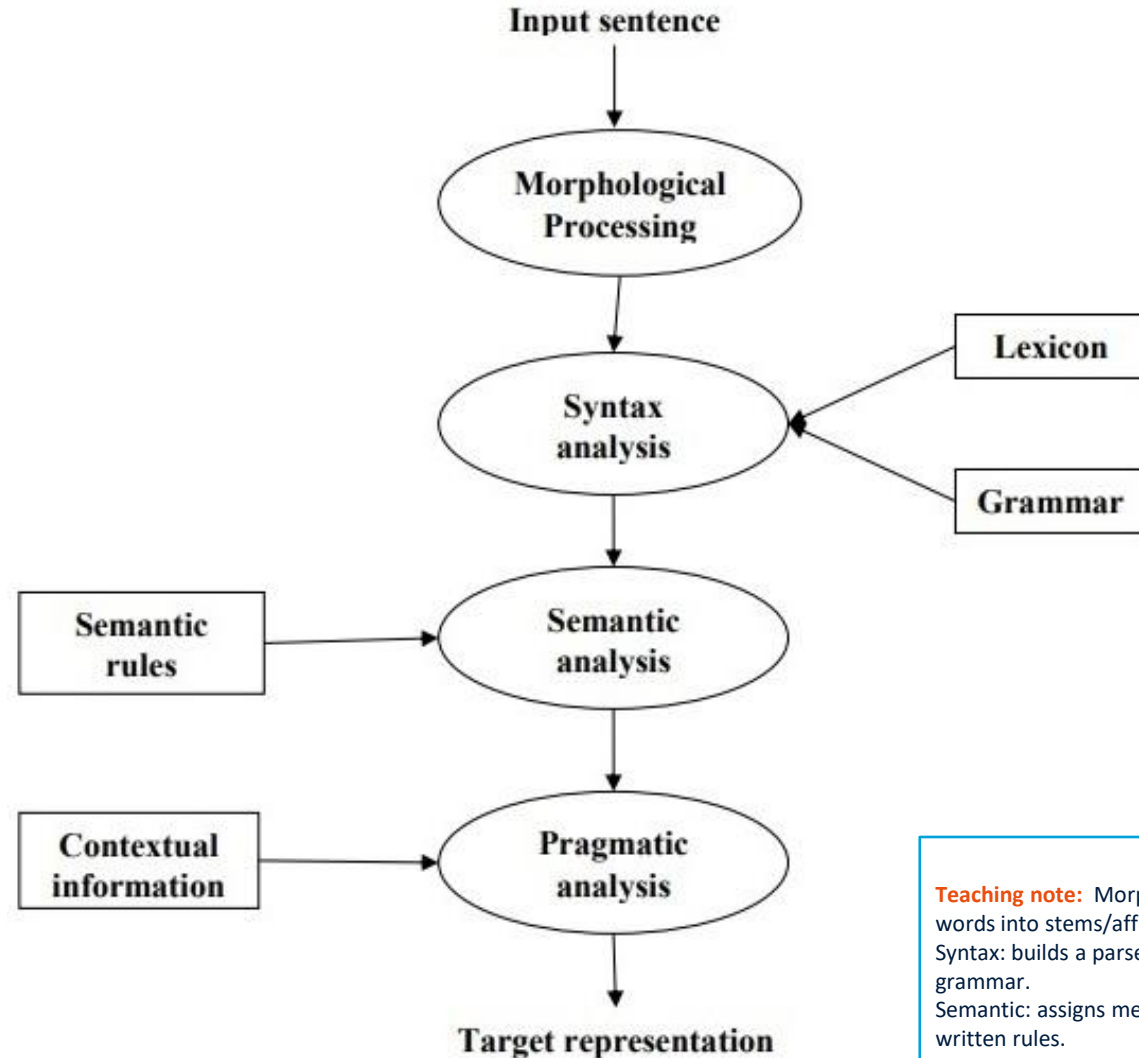
Natural Language Processing: some definitions

- Words: morphology (lexicon or vocabulary)
- Sentences: syntax (grammar)
- Meaning: semantics
- Conversation: discourse, intent & goals



Architecture of first-generation NLP systems

1970-1990



Teaching note: Morphological: splits words into stems/affixes.
Syntax: builds a parse tree via lexicon + grammar.
Semantic: assigns meaning via hand-written rules.
Pragmatic: uses context to resolve intended meaning (e.g. coreference).

Architecture of first-generation NLP systems

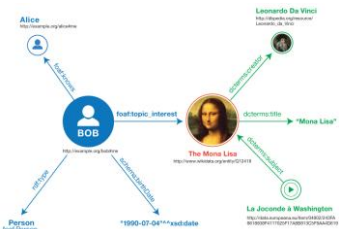
1970-1990

$$\frac{[Q[e/E]] \quad x := E \quad [Q]}{[P \wedge B] \quad C \quad [R] \quad [P \wedge \neg B] \quad D \quad [R]} \\ [P] \text{ if } B \text{ then } C \text{ else } D \quad [R]$$

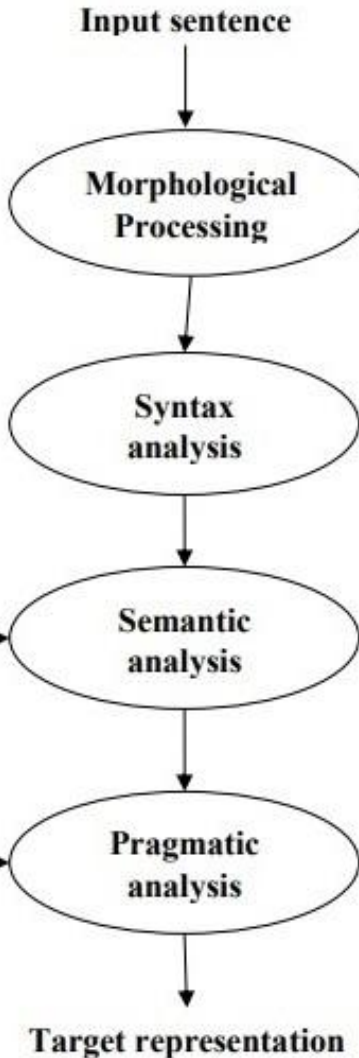
$$\frac{[I \wedge B \wedge (V = Z)] \quad C \quad [I \wedge (V < Z)], \quad I \wedge (V \leq 0) \Rightarrow \neg B}{[I] \text{ while } B \text{ do } C \quad [I \wedge \neg B]}$$

$$\frac{P \Rightarrow P' \quad [P'] \quad C \quad [Q'] \quad Q' \Rightarrow Q}{[P] \quad C \quad [Q]}$$

Various types of logic (1st, 2nd order, logic of argumentation, ...)



Resource Description Framework (RDF) or Ontologies



Abstraction: fielded records

key	field	field	field	field
key	field	field	field	field

Eg: dictionary

wake: weik, [v], cease to sleep...
walk: wa:k, [v], progress by lifting and setting down each foot...

Eg: comparative wordlist

wake; aufwecken; acordar
walk; gehen; andar
write; schreiben; enscrever

Eg: verb paradigm

wake	woke	woken
write	wrote	written
wring	wrung	wrung

Lookup Tables

$S \rightarrow \langle NP PP \rangle$
 $NP \rightarrow [NP VP]$
 $NP \rightarrow [DT NP]$
 $VP \rightarrow [VP VP]$
 $VP \rightarrow [TO VP]$
 $VP \rightarrow [VP NP]$
 $NP \rightarrow [JJ NN]$
 $PP \rightarrow \langle IN NP \rangle$
 $NP \rightarrow [PRP\$ NN]$

Various types of Grammars

Annotated Linguistic Corpora

“Text (in its natural context) annotated with syntactic & semantic information”. Early form of Machine Learning.

Examples of text corpora for computational linguistics:

- Brown corpus
- Penn Tree bank
- Google books ngram corpus
- GUM (Georgetown) corpus
- ...

See also:

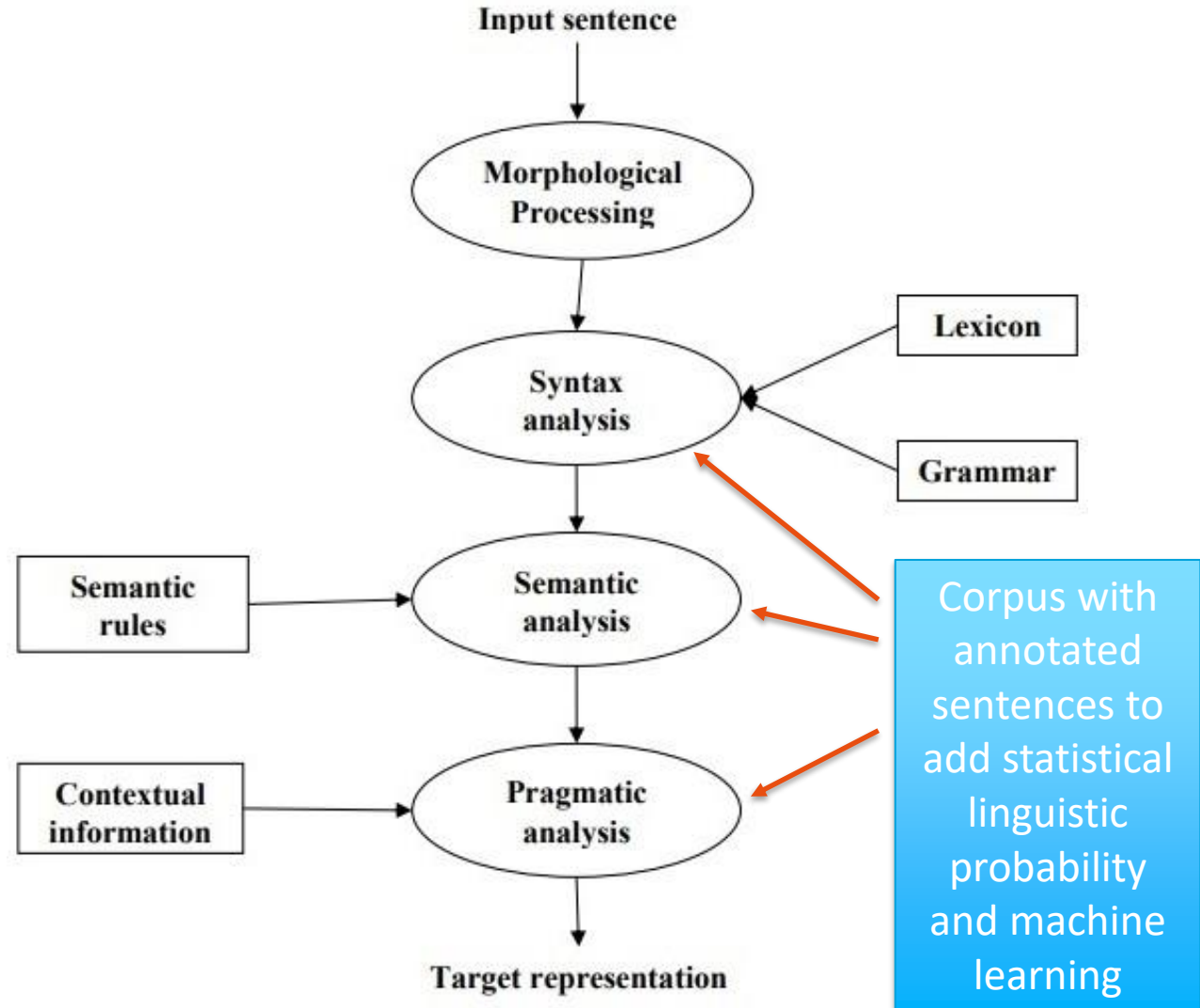
https://en.wikipedia.org/wiki/List_of_text_corpora

The screenshot shows a web application titled "tk treebank search #60". It has a search interface with fields for "Sentence File" (lu.lisp) and "Prolog Tree File" (lu.pl), and a "Load Files" button. Below these are checkboxes for "Match Sentence" (TO) and "Match Tree (Prolog)" (node(X,VP).branching(X,3)). The "Sentence Count" is 317, and "Selected" is 104. The "Show" button is set to "Selected Only". The "Displayed Tree (Sentence)" is 43. The main content area displays a list of sentences, each with its constituent parts in parentheses. The sentence "((I (PRP)) (am (VBP)) (eager (JJR)) (to (TO)) (be (VB)) (her (PRP)))" is highlighted. To the right of the list is a syntax tree for the sentence "It is illegal for John to park here". The tree structure is as follows: S branches into NP (PRP: It) and VP (VBZ: is). VP branches into ADJP (JJ: illegal) and SBAR (IN: for). SBAR branches into S, which branches into NP (NNP: John) and VP (VP: to). The inner VP branches into VB (park) and ADVP (RB: here).

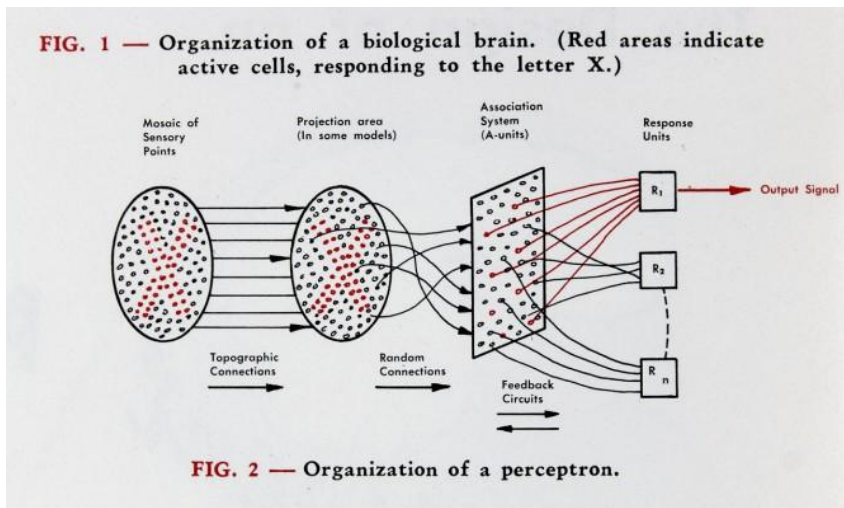
Architecture of second-generation NLP systems

1985-2010

Teaching note: Same 4 pipeline boxes as before — now each is also informed by corpus statistics, not only hand-written rules. Evolution, not replacement.



Parallel to traditional AI, there has always been interested in Artificial Neural Networks & Language.



1957. Frank Rosenblatt had a vision that he could make computers see and understand language using Perceptrons.

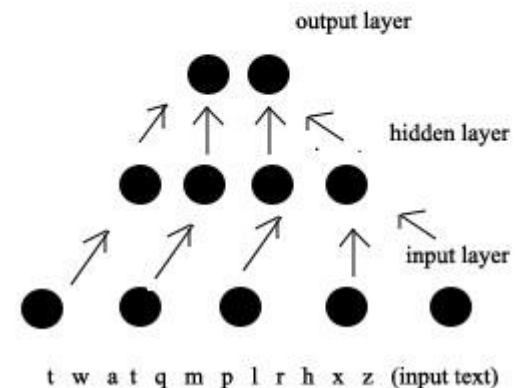
→ first trainable neuron: linear boundaries only.



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teacher: /t/ /w/ /a/ /t/

Guess: /t/ /a/ /w/ /x/

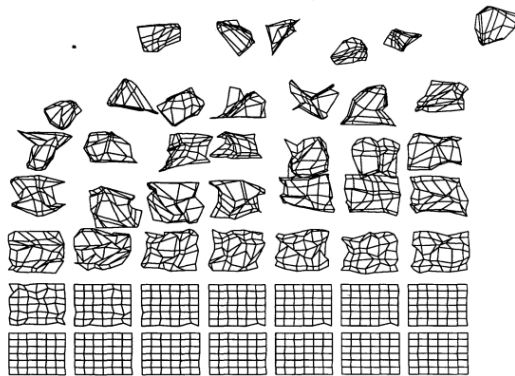


Mid 1980's: impressive speech synthesis NETTalk by Terrence Senowski using back propagation.

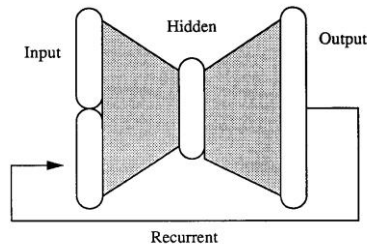
→ proof a network could learn language, not be hand-coded.

1990-1997: first attempts with back propagation, Recurrent Neural Networks and self-organizing maps.

Neural Networks in Natural Language Processing and Information Retrieval



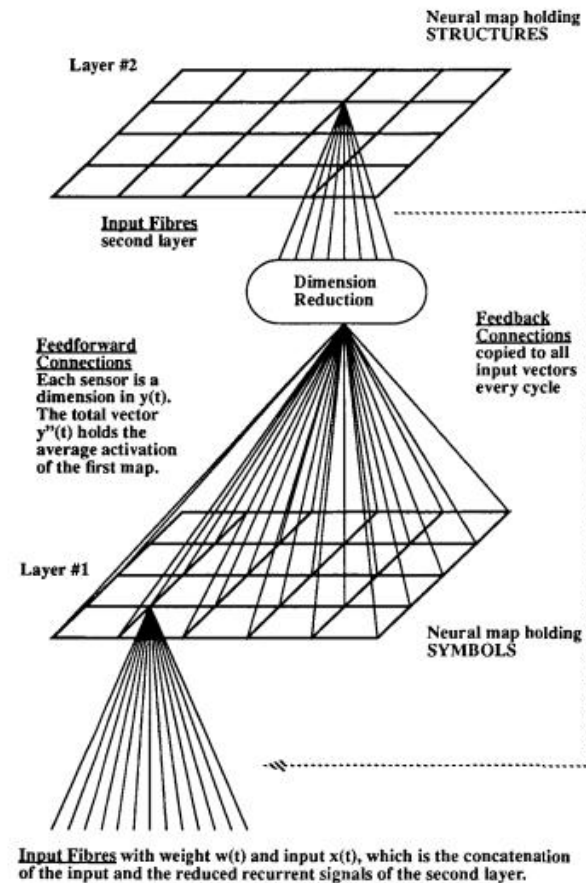
Johannes Cornelis Scholtes



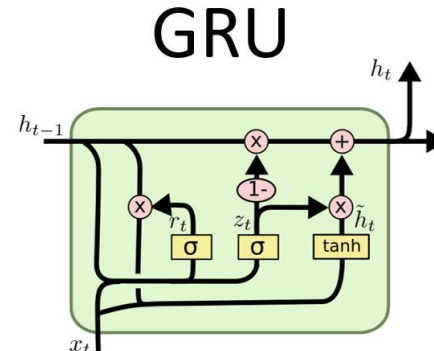
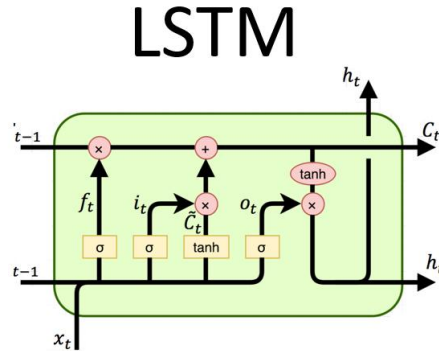
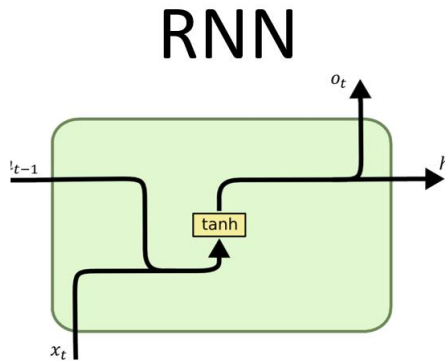
Challenges:

- Not enough data
- Not enough CPU power
- Not enough memory
- Not able to get multi-layer training stable with then known algorithms.

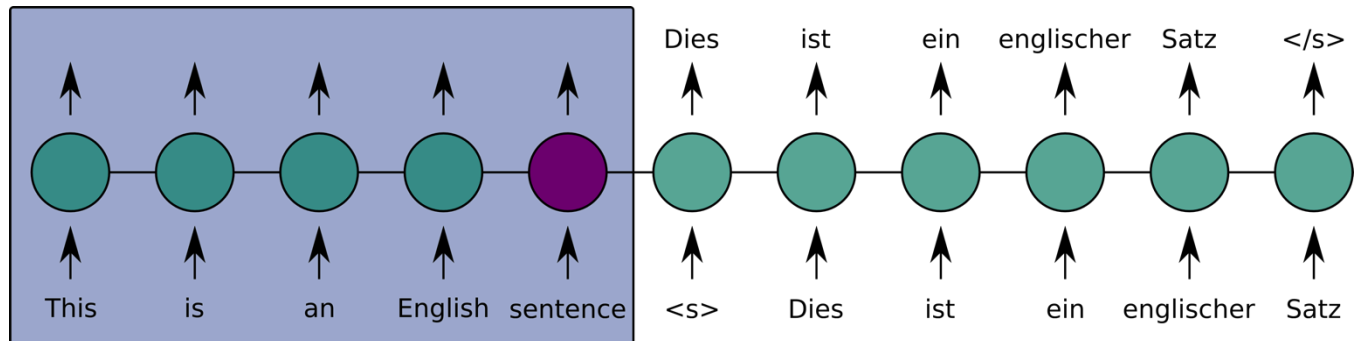
These 4 constraints are exactly what changed by 2012–2017, enabling deep learning.



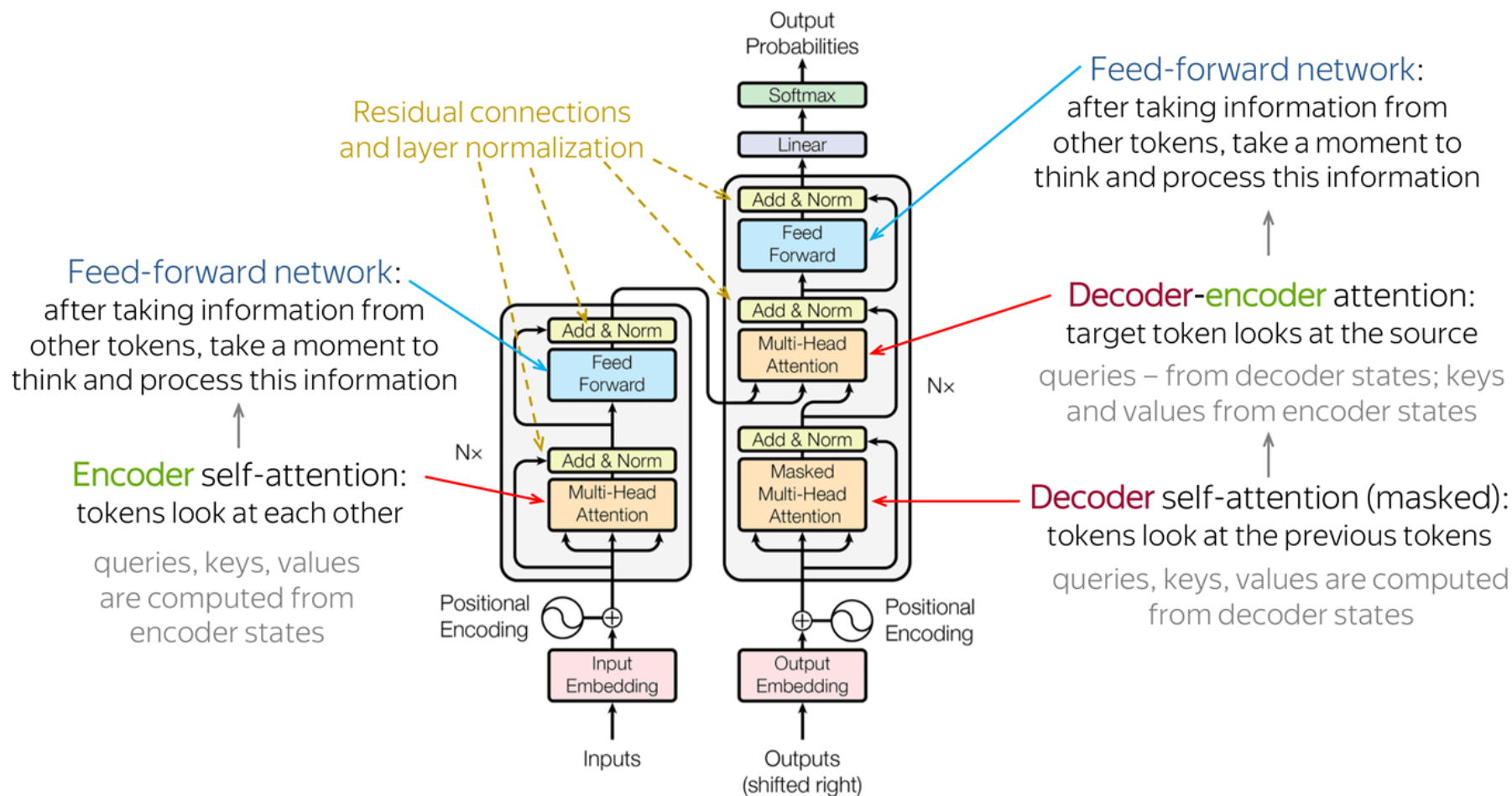
1997-2017: RNN, LSTM, and GRU ...



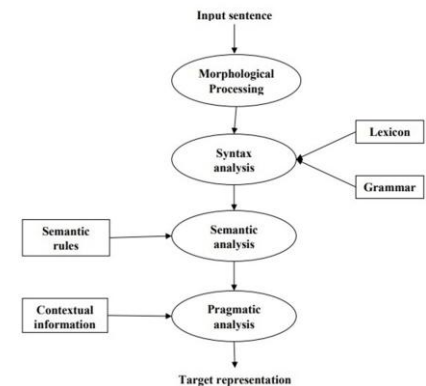
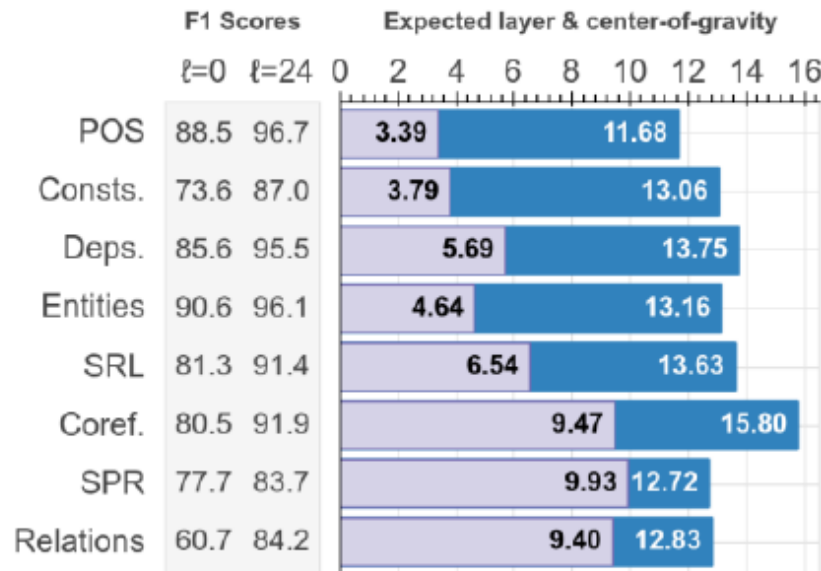
Teaching note: RNN: forgets over long sentences.
LSTM: gates decide what to keep.
GRU: simplified LSTM.
All 3 are sequential — exactly what Transformers remove next.



2017: Transformers – SOTA for all NLP tasks



Different NLP Tasks in BERT (Transformer encoder-only model) Layers: rediscovering the Classical NLP pipeline!



BERT layers vs NLP pipeline sub-tasks [Tenney et al., 2019]

POS = Part of Speech Tagging (noun Verb, Determiner, ...)

Consts = Constituency Parsing (NP, VP)

Deps = Dependency Parsing

Entities = Named Entity Extraction

SRL = Semantic Role Labeling

Coref = Coreference and Anaphora resolutions

SPR = SPR data set for Semantic and Prototypes Relations

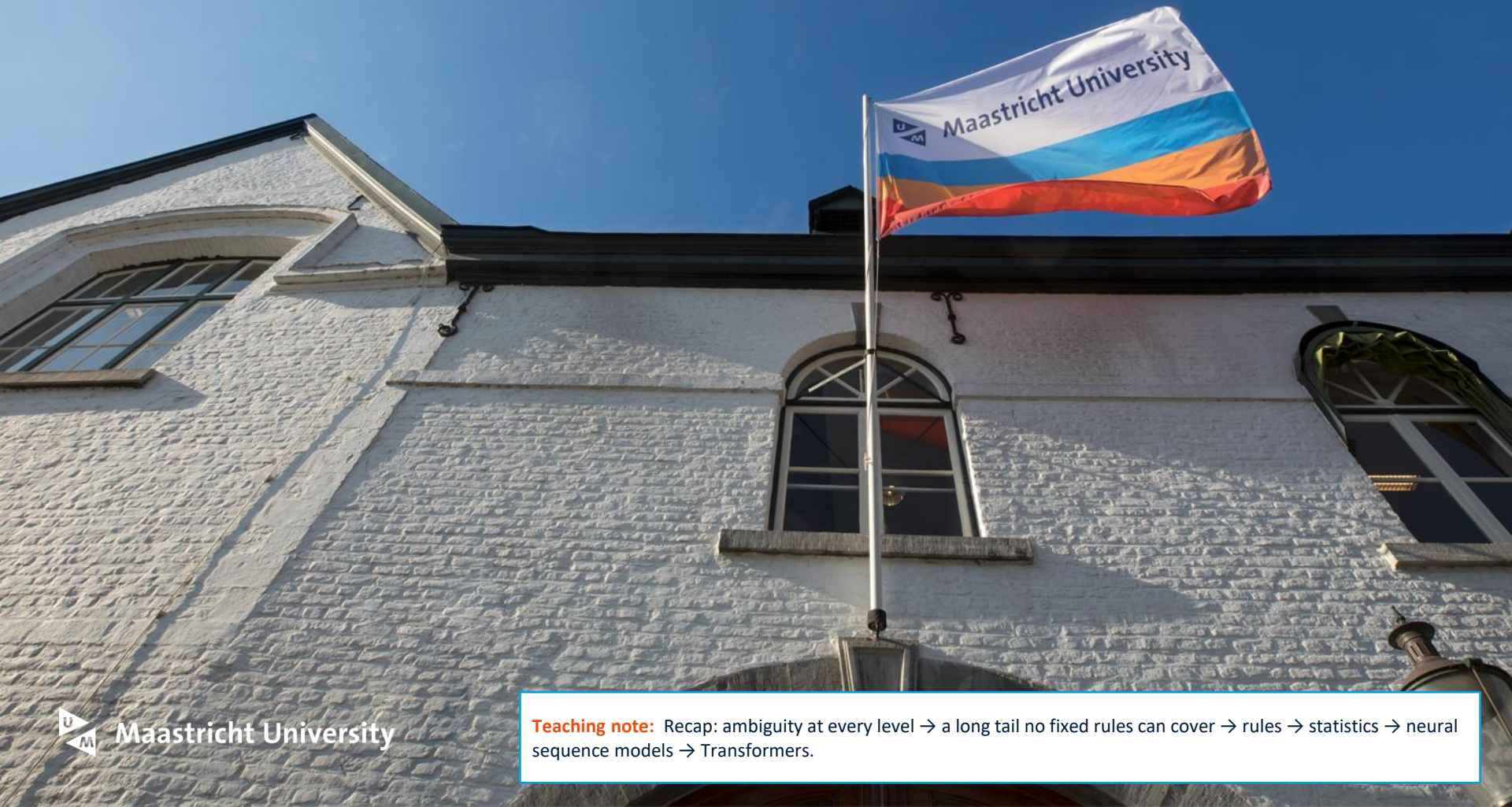
Relations = Complex relations (RDF – Subject – Predicate – Object)

Tenney, I., Das, D., and Pavlick, E. (2019). Bert rediscovers the classical nlp pipeline. arXiv preprint arXiv:1905.05950.

XAI & Linguistics

- XAI = eXplainable AI.
- For proper XAI, it really helps to understand and investigate traditional linguistic phenomena in Deep Learning Models.
- These will help one to understand, explain and ultimately resolve errors.

Questions?



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Teaching note: Recap: ambiguity at every level → a long tail no fixed rules can cover → rules → statistics → neural sequence models → Transformers.

Exercise 1

Take the sentence: *"We saw the flight cancelation notice using a borrowed laptop."*

- Identify all distinct interpretations stemming from syntactic ambiguity (specifically prepositional phrase attachment).
- Classify where the ambiguity originates according to the linguistic levels discussed in lecture: lexical, syntactic, or semantic.

Exercise 2

Explain how Zipf's law ($f \propto \frac{1}{r}$) creates a failure mode for both rule-based and machine-learning NLP systems when processing modern informal text (such as social media slang, OCR errors, or neologisms like "*Apple boterkoek ge-update*" or "*alter eagle*").

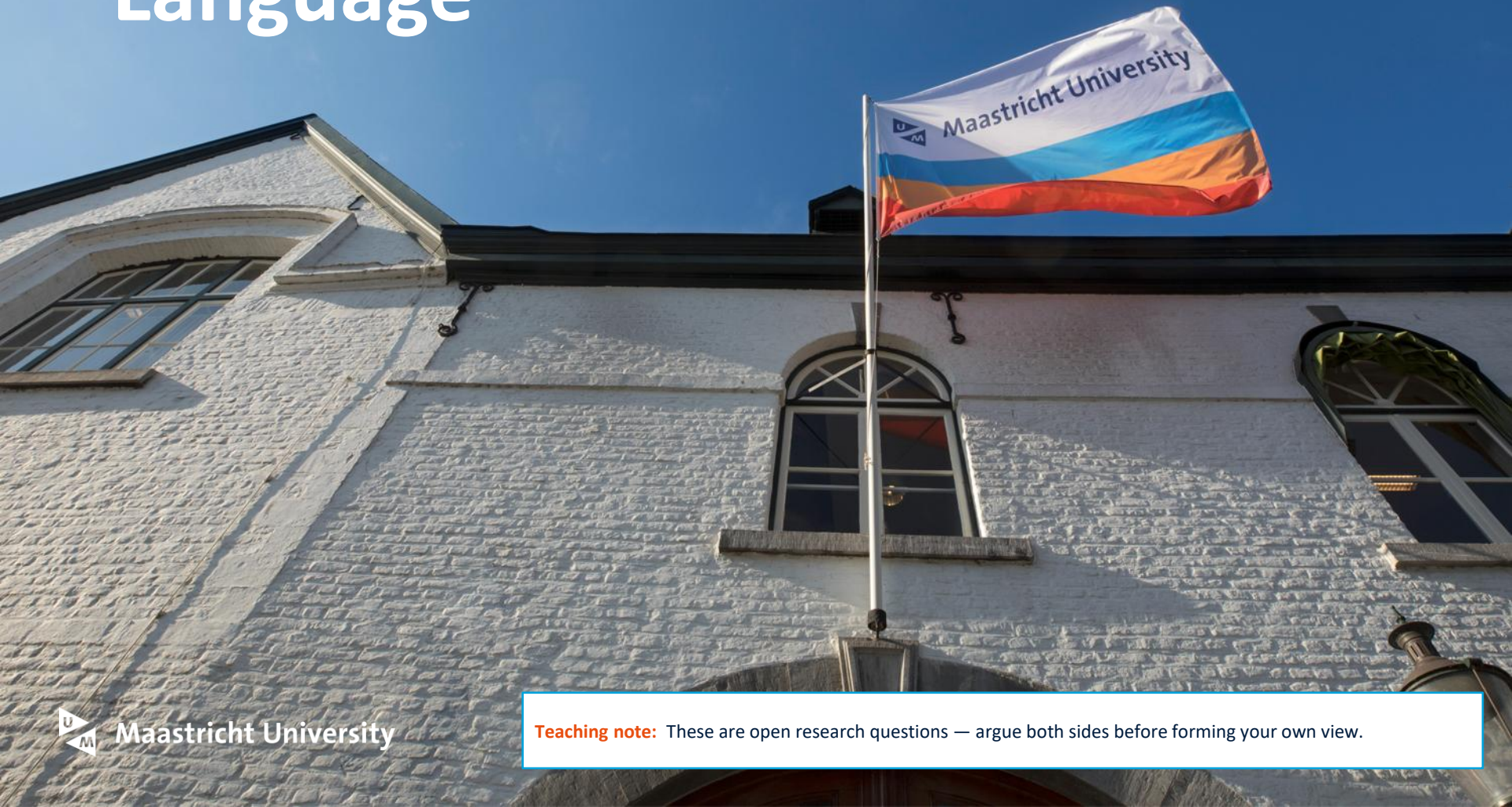
- Why does the mathematical reality of a "long tail" make pure rule-based grammars and lookup dictionaries continuously incomplete?
- What failure mode occurs if a statistical or deep learning model encounters tokens that sit far out on the extreme end of the tail?

Exercise 3

This lecture illustrates that BERT layers "rediscover" the classical NLP pipeline.

- Look at the center-of-gravity chart for tasks such as POS tagging, dependency parsing, coreference resolution, and relation extraction. Which tasks peak in the early layers, and which require the higher layers?
- Why would a self-attention mechanism naturally solve local syntactic dependencies before it can resolve coreference or complex semantic relations?
- Compared to traditional NLP, where syntax, semantics, etc are separated in clear different parts of the analysis, here we see a more fluid border between the different linguistic notions. What can you conclude from this?

Discussion Lecture: The Problem with Natural Language



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Teaching note: These are open research questions — argue both sides before forming your own view.

Resolving Linguistic Ambiguity, World Knowledge and Intent

In the sentence “If the baby does not like the milk, boil it,” humans immediately resolve the pronoun “it” to the milk rather than the baby.

- Does disambiguating this rely on language structure (syntax/lexicon), or does it require knowledge of how the physical world works?
- If an LLM or neural model correctly resolves it, can we claim the model truly ‘understands’ what milk and babies are, or has it simply learned co-occurrence statistics from text?

On the Long Tail, Zipf's Law, and Rule-Based vs. Statistical NLP

We saw that language exhibits a strict 'long tail' driven by Zipf's law, making rule-based grammar engineering fragile because rare expressions, slang, and neologisms (like 'Brexit' or social media errors) are perpetually omitted.

- However, since rare tokens in the long tail appear infrequently in training corpora, aren't purely statistical and deep learning models equally vulnerable to this phenomenon?
- If both approaches struggle with the tail, what makes a modern neural architecture fundamentally better at generalising to unseen language than a formal grammar?

On the "Accidental" Emergence of the Linguistic Pipeline in Deep Learning

Tenney et al. (2019) demonstrated that BERT's layers rediscover the classical NLP pipeline—morphology and POS tags resolve early, syntax in the middle, and semantic/coreference relations in the higher layers.

- If an end-to-end transformer trained strictly on next-token/masked-token prediction organically organizes itself to mirror 1980s modular linguist-designed pipelines, does this prove that the classical Chomskyan linguistic hierarchy is an objective, universal property of language, or is it simply an artifact of how computational systems must compress hierarchical data?

More References

- Turing, A.M., 1950. Computing Machinery and Intelligence. *Mind*, 59(236), pp. 433–460.
- Searle, John (1984), *Minds, Brains and Science: The 1984 Reith Lectures*, Harvard University Press, ISBN 978-0-674-57631-5 paperback: ISBN 0-674-57633-0.